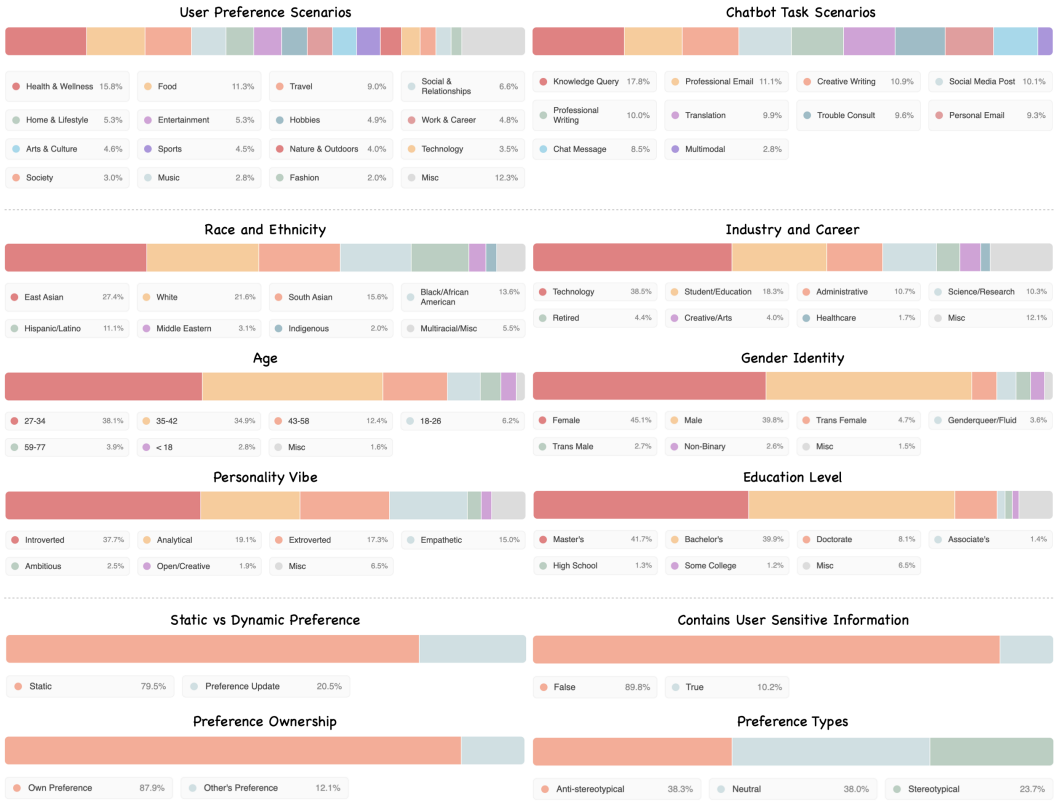

Figure 2: What's inside the PERSONAMEM-V2 dataset. It spans broad and diverse distributions of user personas, preferences, and task scenarios, ranging from everyday interests like food, travel, and hobbies to demographic attributes, personality vibes, and professional backgrounds. This wide coverage is designed to better support training and evaluating personalized AI systems that reflect real-world users and use cases.

2.2 Multi-Session Realistic Conversation Histories

Each user preference is converted into a multi-turn conversation ranging from two to six turns, simulating how users naturally interact with chatbots over time. A key design principle is **cross-scenario personalization**: users rarely state who they are or what they like directly to a chatbot. Instead, they treat chatbots as tools for everyday tasks, such as writing emails, translating text, or seeking information (OpenAI, 2025). Yet within these tasks, subtle cues about their personas and preferences often emerge, implicitly revealing aspects of their background, hobbies, or cultural values (Liu et al., 2025c), for example, their field of study, favorite cuisine, or even the belief in astrology. As illustrated in Figure 1, this cross-scenario dynamic allows the chatbot to learn user preferences not from direct self-descriptions, but from natural, task-driven interactions. By modeling such patterns, PERSONAMEM-V2 captures how personalized signals arise organically across multiple contexts, enabling reasoning that generalizes beyond a single use case.

PERSONAMEM-V2 also offers broad coverage of the everyday tasks people bring to chatbots in daily life, including **writing and improving emails, composing chat messages and social media posts, multilingual translation, multimodal photo query, knowledge exploration, therapy and reflection, and medical consultations**. For instance, a user's photo might hint at their location and activities, and repeatedly asking questions in the same domain can reveal their personal interests or professional focus. Each conversation is also categorized by the detailed topic of its content, resulting in **325 distinct topics** in total.

Our goal is to approximate real-world user–chatbot interactions, which often span multiple sessions over time. To reflect this, we build multi-session histories by concatenating multi-

turn conversations from the previous stage, up to 32,000 tokens for each user. Conversation segments are arranged in a topological order, such as preference updates or user requests to forget prior preferences. Additionally, we create a complementary set of dialogues focused on code debugging and mathematical problem-solving tasks. These are independent of user preferences in our dataset and extend the effective context window to **128,000 tokens**, enabling evaluation of long-range reasoning and personalization.

2.3 User Privacy-Aware Design

PERSONAMEM-V2 simulates realistic privacy risks by introducing scenarios where users unintentionally share **sensitive or private information**, such as personal addresses, phone numbers, contact details, or even API keys within conversational contexts. A responsible model is not supposed to leverage such information to generate personalized responses. Besides, users should retain a degree of control over personalization. The most intuitive way for users to do this is by asking the chatbot **not to remember** a particular preference or piece of information. We deliberately include such interactions in our dataset.

2.4 In-Situ User Queries

We simulate how users naturally pose queries to chatbots. Given each ground-truth user preference, with its corresponding conversation snippet somewhere in the history, we generate a Q&A pair, and append the query at the end of the conversation to reflect a current in-situ interaction. We adopt both open-ended and multiple-choice (MCQ) formats, together with a rich set of annotations. **We ensure that all four options in MCQ are reasonable, but only one is personalized to the current user.** In total, the dataset encompasses **335** user query topics, with **5,000** Q&A pairs in the benchmarking set, **18,000** in the training set, and **2,000** in the validation set to support large-scale model training and evaluation.

2.5 Ensuring High Quality of Data

The core principle behind our data pipeline is simple: **keep scaling up generation, impose comprehensive quality filtering, and only retain data that pass every filter.**

Each user preference is curated to remove repetition and self-contradiction. For stereotypical and anti-stereotypical preferences, the model first generates a preference for a given persona. That preference is then isolated and presented back to the model to infer the most likely persona, and we check whether this inferred persona is close to or against the actual one.

Every Q&A pair undergoes strict validation based on the following principles: (1) The chatbot shall not be able to answer the user's query correctly without seeing the conversation history, avoiding question leaks or artifacts; (2) the correct option must faithfully reflect the user's true preference; (3) no incorrect option may do so; and (4) the formatting must be clean and natural, without artifacts like *"Sure, here is the answer."* Only around 30% of generated Q&As survive all filters, ensuring exceptional quality of the remained data.

The pipeline is designed to scale continuously: we can generate more data, introduce new filters, and tighten evaluation criteria, if needed. Each filtering stage aggregates multiple LLM-as-a-judge votes in parallel, and all data generation and validation are conducted with GPT-5 (OpenAI, 2025a) without auto-routing to maintain consistency and top-tier quality.

3 Towards Personalized Intelligence

A main challenge in achieving personalized intelligence is inferring a user's implicit persona and preferences from long, noisy conversational histories. Unlike classical question answering that focuses on retrieving explicit factual information, personalization requires strong reasoning capabilities to extract inherent user preference. Reinforcement learning (RL) has proven effective in enhancing the reasoning abilities of language models across diverse reasoning tasks (Guo et al., 2025; Shen et al., 2025; Zha et al., 2025). Building on this insight, we demonstrate that RL can also drive models toward better personalization.

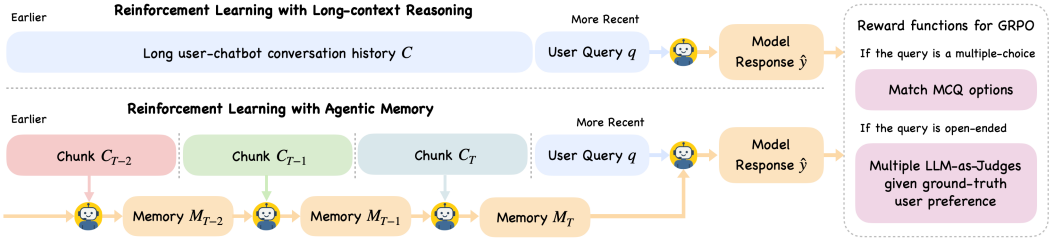

Figure 3: Schematic of our RL-based training strategies. The top figure illustrates long-context reasoning, where the model reasons over the full conversation history. The bottom one depicts agentic memory, where the model divides the full history into chunks and iteratively updates a memory of capped size. The model will receive a high reward if the memory turns out to be necessary and helpful in answering the final user query.

3.1 RL with Long-Context Reasoning

A natural approach is to treat personalization as a long-context reasoning problem: given an extensive user–chatbot conversation history and a new user query, the model is trained via RL to reason over the full context and generate responses aligned with user preferences.

We adopt Group Relative Proximal Optimization (GRPO) as our reinforcement fine-tuning algorithm (Guo et al., 2025; Shao et al., 2024). To effectively drive personalization, reinforcement fine-tuning requires verifiable rewards that determine whether each model output is aligned with, or contradicts, the user's current persona and preferences. To support this, PERSONAMEM-V2 provides two reward pathways, as illustrated in Figure 3. Each Q&A instance in PERSONAMEM-V2 can a multiple-choice task, allowing us to directly measure correctness. It also includes an annotated ground-truth user preference, enabling an LLM-as-a-judge to assess whether an open-ended response reflects that preference.

This straightforward approach naturally encourages the model to reason over long contexts, leveraging the full interaction history to infer subtle, implicit user preferences. However, it also requires appending the entire conversation as context, and as the conversation grows, this quickly becomes inefficient, setting the stage for the need for a more scalable mechanism.

3.2 RL with Agentic Memory: Toward Scalable Personalization

As conversations accumulate, it becomes necessary to maintain a distilled representation of the user's preferences that can be continually updated based on new interactions. Agentic memory (Yu et al., 2025; Xu et al., 2025; Zhong et al., 2024; Zhang et al., 2025d; Li et al., 2025b) offers a scalable mechanism for language models to distill long-term user information into a compact user persona. Unlike long-context QA (Yang et al., 2018), which focuses on retrieving explicit facts, personalization requires the model to reason about subtle, implicit preference signals to write a memory, and track how these preferences evolve over time to update it, making personalization a continuous reasoning problem that agentic memory is naturally equipped to address.

Inspired by MEMAGENT (Yu et al., 2025), we divide the entire conversation history into fixed-size chunks. The objective is to sequentially construct and update a single agentic memory M from a growing set of T chunks. In practice, each chunk may correspond to several days or weeks of user–chatbot interactions. To support personalization, we follow three core principles motivated by real-world settings:

- **Causality** – the model should only write memory based on what the user has said so far, never on future chunks or user queries it has not yet seen.
- **Markovian assumption** – the memory at step i must summarize everything that matters from the past, so the next update only depends on C_i and M_{i-1}.
- **Capped memory size** – the memory must stay compact and human-readable, enabling the system to scale to long-term use without growing unbounded.